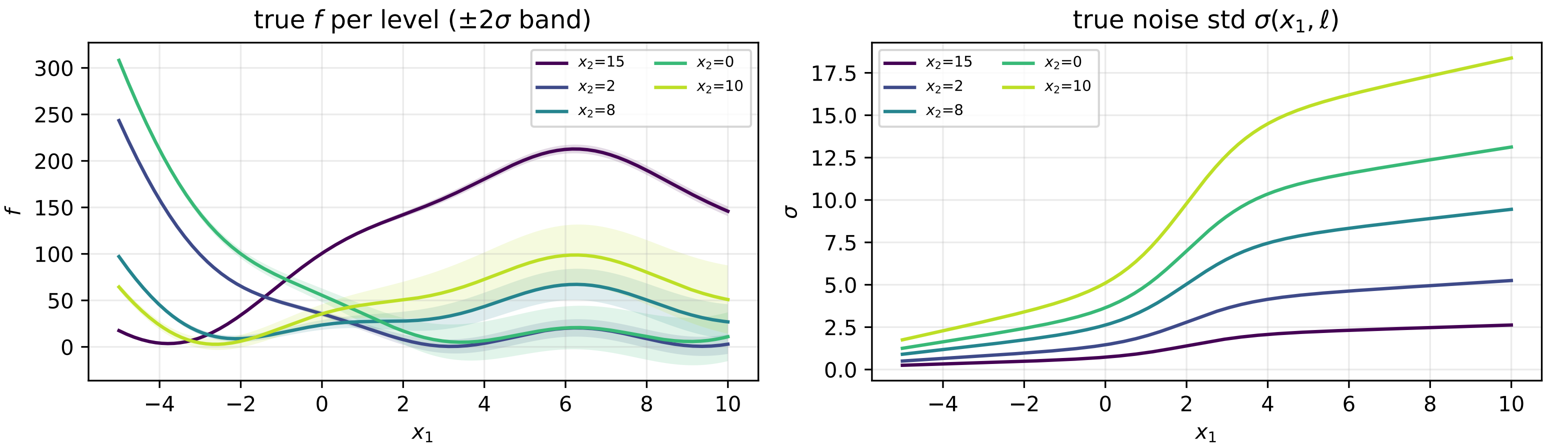
Test problems

Branin (heteroscedastic, 5 levels)

$$f(x_1,\ell) = \left(x_2(\ell) - \frac{5.1}{4\pi^2}x_1^2 + \frac{5}{\pi}x_1 - 6\right)^2 + 10\left(1 - \frac{1}{8\pi}\right)\cos x_1 + 10, \quad x_1 \in [-5,10],\ x_2(\ell) \in \{15,2,8,0,10\}$$

$$\sigma(x_1,\ell) = \left[0.5 + 0.15(x_1+5) + \frac{2.5}{1+e^{-1.2(x_1-2)}}\right] m(\ell), \quad m(\ell) \in \{0.5,1.0,1.8,2.5,3.5\}$$

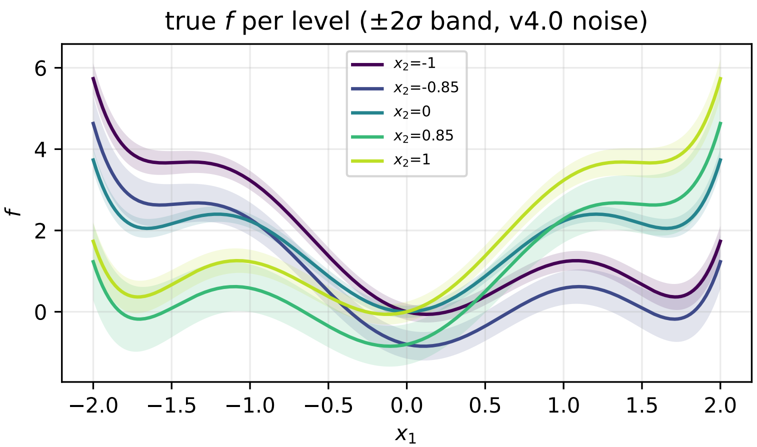


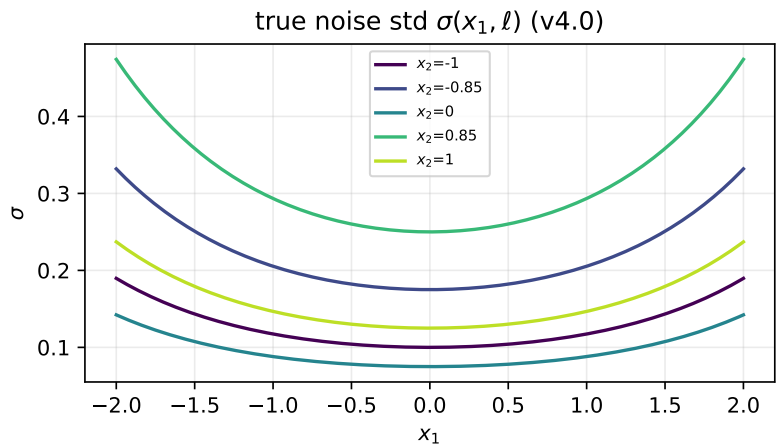
Six-hump camel (5 levels: two pairs + bridge)

$$f(x_1,\ell) = \left(4 - 2.1x_1^2 + \frac{x_1^4}{3}\right)x_1^2 + x_1 x_2(\ell) + (-4 + 4x_2(\ell)^2)x_2(\ell)^2, \quad x_1 \in [-2,2],\ x_2(\ell) \in \{-1.0,-0.85,0.0,0.85,1.0\}$$

$$\sigma(x_1,\ell) = 0.05\, e^{(0.4x_1)^2}\, m(\ell), \quad m(\ell) \in \{2.0,3.5,1.5,5.0,2.5\}$$

(noise as used in the 30-seed replication tables below)

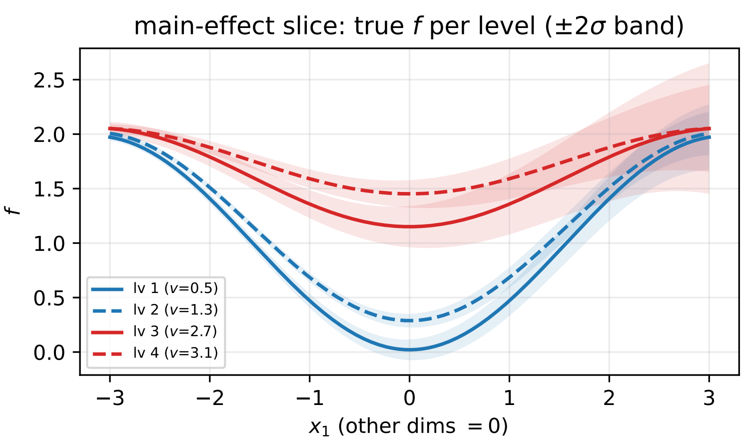




Griewank 6-D (level-shifted pairs)

$$f(\mathbf{x}_q, \ell) = \sum_{i=1}^{6} \frac{xs_i^2}{4000} - \prod_{i=1}^{6} \cos\left(\frac{xs_i}{\sqrt{i}}\right) + 1 + b(\ell), \quad xs = [\mathbf{x}_q, v(\ell)],\ \mathbf{x}_q \in [-3, 3]^5$$

$$v(\ell) \in \{0.5, 1.3, 2.7, 3.1\}, \quad b(\ell) \in \{0, 0.15, 0.6, 0.75\}, \quad \sigma(\mathbf{x}_q, \ell) = 0.01 \cdot 10^{(x_1+3)/6}\, m(\ell), \quad m(\ell) \in \{1.5, 1.0, 3.0, 2.0\}$$



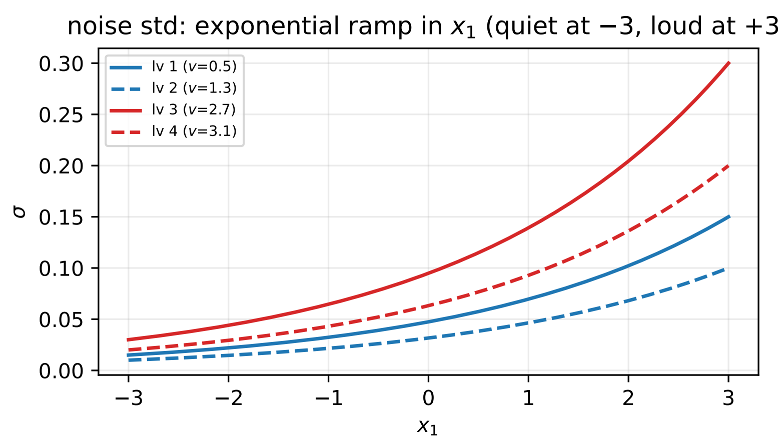


Table 1: 30-seed replication, 1-D problems (fixed design, fresh noise replicates per seed): mean ± std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95).

Problem	n	Model	MAE	RRMSE	IS	Coverage
Branin	25	Separate GPs	10.91 ± 0.466	0.434 ± 0.022	113.7 ± 11.60	0.775 ± 0.028
		Categorical GP	4.529 ± 0.241	0.159 ± 0.011	38.87 ± 2.018	**0.946 ± 0.010**
		Standard LVGP	4.419 ± 1.063	0.212 ± 0.051	74.80 ± 30.69	0.442 ± 0.143
		H-LVGP	**2.520 ± 0.723**	**0.096 ± 0.037**	**37.28 ± 20.11**	0.996 ± 0.016
	40	Separate GPs	4.454 ± 0.361	0.208 ± 0.017	42.88 ± 1.784	**0.931 ± 0.011**
		Categorical GP	**1.585 ± 0.268**	**0.054 ± 0.012**	**14.86 ± 0.896**	0.996 ± 0.010
		Standard LVGP	3.109 ± 0.705	0.122 ± 0.030	62.82 ± 20.09	0.410 ± 0.145
		H-LVGP	1.722 ± 0.425	0.064 ± 0.019	35.61 ± 10.83	0.998 ± 0.008
Six-hump camel	25	Separate GPs	0.231 ± 0.009	0.319 ± 0.007	2.072 ± 0.061	**0.952 ± 0.008**
		Categorical GP	0.168 ± 0.009	0.262 ± 0.010	**1.778 ± 0.136**	0.923 ± 0.012
		Standard LVGP	**0.152 ± 0.018**	**0.242 ± 0.029**	3.405 ± 0.521	0.417 ± 0.097
		H-LVGP	0.178 ± 0.012	0.287 ± 0.016	2.153 ± 0.231	0.905 ± 0.017
	40	Separate GPs	0.196 ± 0.008	0.295 ± 0.006	2.313 ± 0.111	0.888 ± 0.015
		Categorical GP	**0.114 ± 0.010**	**0.160 ± 0.015**	**1.127 ± 0.164**	0.918 ± 0.013
		Standard LVGP	0.145 ± 0.034	0.212 ± 0.066	3.240 ± 1.092	0.324 ± 0.106
		H-LVGP	0.143 ± 0.024	0.226 ± 0.042	1.147 ± 0.318	**0.953 ± 0.029**

Table 2: 30-seed replication, high-dimensional problems (fixed design, fresh noise replicates per seed): mean ± std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95).

Problem	n	Model	MAE	RRMSE	IS	Coverage
Griewank 6-D	48	Separate GPs	0.076 ± 0.003	1.086 ± 0.036	0.833 ± 0.034	0.861 ± 0.014
		Categorical GP	**0.066 ± 0.003**	**0.939 ± 0.048**	**0.695 ± 0.027**	**0.893 ± 0.015**
		Standard LVGP	0.104 ± 0.009	1.376 ± 0.134	1.385 ± 0.193	0.716 ± 0.052
		H-LVGP	0.097 ± 0.007	1.242 ± 0.169	0.939 ± 0.070	0.845 ± 0.021
	64	Separate GPs	0.066 ± 0.002	0.973 ± 0.025	0.773 ± 0.031	0.882 ± 0.010
		Categorical GP	**0.064 ± 0.003**	**0.896 ± 0.041**	**0.748 ± 0.022**	**0.912 ± 0.008**
		Standard LVGP	0.086 ± 0.012	1.210 ± 0.168	1.203 ± 0.273	0.736 ± 0.043
		H-LVGP	0.084 ± 0.026	1.141 ± 0.321	0.916 ± 0.226	0.876 ± 0.029
	80	Separate GPs	0.071 ± 0.002	1.016 ± 0.040	0.730 ± 0.028	0.871 ± 0.010
		Categorical GP	**0.061 ± 0.003**	**0.986 ± 0.046**	**0.723 ± 0.024**	**0.887 ± 0.009**
		Standard LVGP	0.100 ± 0.010	1.359 ± 0.145	1.084 ± 0.228	0.820 ± 0.047
		H-LVGP	0.082 ± 0.015	1.133 ± 0.277	0.950 ± 0.089	0.836 ± 0.047
	120	Separate GPs	0.064 ± 0.003	0.903 ± 0.036	0.690 ± 0.063	0.880 ± 0.017
		Categorical GP	**0.050 ± 0.001**	**0.749 ± 0.020**	**0.539 ± 0.015**	**0.922 ± 0.008**
		Standard LVGP	0.101 ± 0.015	1.382 ± 0.186	1.283 ± 0.269	0.773 ± 0.048
		H-LVGP	0.111 ± 0.033	1.564 ± 0.605	1.101 ± 0.182	0.856 ± 0.031
	160	Separate GPs	0.056 ± 0.002	0.851 ± 0.040	0.683 ± 0.052	0.836 ± 0.018
		Categorical GP	**0.047 ± 0.003**	**0.704 ± 0.053**	**0.598 ± 0.024**	0.886 ± 0.011
		Standard LVGP	0.075 ± 0.008	1.048 ± 0.107	0.948 ± 0.391	0.811 ± 0.100
		H-LVGP	0.137 ± 0.027	2.091 ± 0.433	1.047 ± 0.132	**0.934 ± 0.020**

Table 3: 30-seed replication, 1-D problems (fixed design, fresh noise replicates per seed): mean ± std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95). Without the Categorical GP (3-model comparison; winners recomputed).

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Table 4: 30-seed replication, high-dimensional problems (fixed design, fresh noise replicates per seed): mean ± std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95). Without the Categorical GP (3-model comparison; winners recomputed).

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